Samantha Scott

Computer Engineering Student • Python & Data Analysis • AI/ML (In Progress) • Azure Certified
samantha_p_scott@yahoo.com • (305) 490-4969 • linkedin.com/in/engineer-samanthascott • github.com/samanthapscott

SUMMARY

Computer Engineering senior at FIU (GPA: 4.0) with hands-on experience in **Python, data analysis, and data visualization** using pandas and matplotlib. Skilled in generating and processing structured datasets, performing **descriptive statistics, data quality control**, and exporting results for research use. Familiar with **GitHub** for code archiving and documentation. Strong interest in **AI/ML applications** with Machine Learning coursework planned for Fall 2026. Seeking a Research Assistant role in the **Neuroinformatics and Brain Connectivity Lab** to contribute to data analysis, literature curation, and AI/ML-driven research.

RESEARCH INTERESTS

Data Analysis & Visualization, Neuroinformatics, AI/ML Applications, Literature Meta-Analysis, Topic Modeling, Brain Connectivity Research

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Assembly
Data & Analysis: Pandas, NumPy, Matplotlib, Data Visualization, Descriptive Statistics, Data Quality Control
AI/ML & Research: Edge Impulse (TinyML), Machine Learning (in progress), Meta-Analysis, Topic Modeling
Tools & Development: Git / GitHub, wxPython, Microsoft Azure (AZ-900), SCCM, JAMF, Active Directory
Systems & OS: Linux, Windows, IoT Sensor Programming, Cloud Data Analytics

RELEVANT PROJECTS

IoT Data Generator & Analysis Application | *Python, wxPython, Pandas, Matplotlib, Faker* *Spring 2025*
- Generated and processed synthetic IoT sensor datasets; performed descriptive statistics and data validation using pandas
- Implemented data visualization (histograms, trend plots) to analyze temperature and humidity time-series patterns
- Exported datasets to CSV and JSON for downstream research use; archived all project code on GitHub
- Applied data quality checks to identify anomalies and ensure dataset integrity before analysis and dissemination

Hand Gesture Recognition System | *Arduino Nano 33 BLE Sense, TinyML, Edge Impulse, C* *Fall 2025*
- Curated and annotated sensor datasets used to train and validate a machine learning classification model
- Analyzed model performance metrics to identify consensus across gesture classes; documented results and methodology
- Maintained organized project files, code archives on GitHub, and records of results throughout the project lifecycle

Traffic Light Controller with Pedestrian Interface | *Zybo Z7 FPGA, Embedded C, VHDL* *April 2026*
- Designed FSM-based real-time control system; recorded project notes and results for ongoing documentation
- Maintained structured project files and code repository to support iterative development and reporting

PROFESSIONAL EXPERIENCE

Student Assistant – IT | Florida International University • Miami, FL *Oct 2024 – Present*
- Performed data quality control and system compliance checks across 100+ enterprise endpoints (SCCM, JAMF, Active Directory)
- Documented and archived technical findings; maintained organized records and project notes for each support case
- Resolved 10–12 weekly technical support tickets at 100% resolution rate; maintained system reliability across residential buildings
- Maintained digital signage infrastructure ensuring accurate real-time content delivery across multiple campus buildings

Student Assistant – Dean of Student Services | Miami Dade College • Miami, FL *Nov 2022 – Apr 2024*
- Managed departmental records and official correspondence; organized and maintained project materials and filing systems
- Assisted in preparation of reports and presentations; coordinated campus events while maintaining a full academic course load

Customer Service Officer | Bank of Nova Scotia Jamaica Ltd • Jamaica *Feb 2016 – Aug 2022*
- Promoted to Head Teller; supervised 4 employees, overseeing compliance and daily balancing
- Elected Service Champion for 2 consecutive years; processed 70+ daily transactions with near-zero error rate
- Delivered consistent customer service across in-person and phone channels in a high-compliance financial environment

EDUCATION

Florida International University • Miami, FL *Aug 2024 – Dec 2026*
B.S. in Computer Engineering | GPA: **4.0/4.0**
Relevant Courses: Data Structures, IoT & Sensor Programming with Python, IoT & Analytics with Cloud Services, Systems Programming, Embedded Systems, Signals & Systems, Machine Learning (Planned – Fall 2026)

Miami Dade College • Miami, FL
Associate of Arts | GPA: **4.0/4.0** *Aug 2022 – Apr 2024*

CERTIFICATIONS & HONORS

Microsoft Certified: Azure Fundamentals (AZ-900) • FIU Transfer Merit Scholarship • 2025–2026 Adolfo Henriques Scholarship • 2023 Grahame & Thelma Somerville Scholarship • **Dean's List** — Miami Dade College & FIU (All Semesters) • Society of Women Engineers (SWE) — FIU Chapter Member